

Roll no: 26k-0006

Ques no 2 (Elevator Simulation)

Algorithm:

- 1) Start
- 2) Initialize $\text{Current_Floor} = 0$
- 3) Set Total number of floor requests $N = 5$.
- 4) For $\text{Request_Num} = 1$ to N
 - a) Read Requested_Floor
 - b) IF $\text{Requested_Floor} > \text{Current_Floor}$
PRINT "Moving Up".
ELSE IF $\text{Requested_Floor} < \text{Current_Floor}$
PRINT "Moving Down".
ELSE IF $\text{Requested_Floor} == \text{Current_Floor}$
PRINT "Doors opening".
- 5) update $\text{Current_Floor} = \text{Requested_Floor}$.
- 6) Print the statement that elevator has stopped at new current floor.
END LOOP
- 7) Stop
PAC

Given Data	Processing Required	Output Requirement
• $\text{Current_Floor} = 0$ • $N = 5$ (No. of requests) • Requested-Floor	• Compare Requested-Floor with Current-Floor. • Use greater than, less than, equality to compare • Statement update: $\text{Current_Floor} = \text{Requested_Floor}$.	• Statement "Moving Up", "Moving Down" or "Doors opening" • Print updated location of elevator.

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IPO chart

Input	Processing	Output
Requested-floor	- Initialize Current-Floor = 0 - Compare new data with current data. - Use comparison logic - Update current data to new data.	- Directional text (whether to move up or down) - Current location update confirmation

Pseudocode

BEGIN

DECLARE Current-Floor = 0, N = 5, Counter,
Requested-floor AS INTEGER.

FOR Counter = 1 to N DO

PRINT "Enter Requested floor: ")

INPUT Requested-floor.

IF Requested-floor > Current-Floor THEN

PRINT "Moving Up"

ELSE IF Requested-floor < Current-floor THEN

PRINT "Moving Down"

ELSE

PRINT "Doors Opening"

ENDIF

Current-floor = Requested-floor

PRINT "Elevator is now at floor: ", Current-floor

ENDFOR

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Flowchart:

